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Chapter 1

Setup and Hypotheses

1.1 Standing hypotheses

Definition 1 (Standing hypotheses). In Lean 4 / Mathlib these are:

```
[Group G] [Fintype G] [Field k] [IsAlgClosed k] [Invertible ((Fintype.card G : k))].
```

The invertibility hypothesis is equivalent to `NeZero ((Nat.card G : k))` and to $\text{char}(k) \nmid |G|$; the prototypical case is $k = \mathbb{C}$. We write $|G|$ for `Fintype.card G`.

1.2 Mathlib API for representations

Definition 2 (Representation). `Representation k G V`, defined as the abbreviation `G →* (V → [k] V)`, lives in `Mathlib.RepresentationTheory.Basic`. Throughout the blueprint `ρ : Representation k G V` stands for the action of G on a k -module V .

Definition 3 ($\rho.\text{asModule}$). Each `Representation k G V` has a canonically associated `(MonoidAlgebra k G)-module .asModule`; this is the same carrier V with $k[G]$ -action $(\sum_g a_g g) \cdot v = \sum_g a_g \rho(g)v$. Conversely, any `Module (MonoidAlgebra k G) M` gives a representation `Representation.ofModule M`.

Definition 4 (Subrepresentation). A `Subrepresentation` is a k -submodule of V closed under $\rho(g)$ for every $g \in G$ (file `Mathlib.RepresentationTheory.Subrepresentation`). The map `Subrepresentation.subrepresentationSubmoduleOrderIso` is an order-preserving bijection between `Subrepresentation` and `Submodule (MonoidAlgebra k G) .asModule`, so subrepresentations and $k[G]$ -submodules are interchangeable.

Definition 5 (Intertwining map). `.IntertwiningMap` (file `Mathlib.RepresentationTheory.Intertwining`) is the bundled type of k -linear maps $V \rightarrow W$ that commute with ρ and σ ; it has `FunLike` and `LinearMapClass` instances and a k -module structure. `Representation.IntertwiningMap.equivAlgEnd` identifies `.IntertwiningMap` with the k -algebra of G -equivariant endomorphisms.

Definition 6 (Isomorphism of representations). `.Equiv` is the bundled type of k -linear isomorphisms $V \simeq W$ that intertwine ρ and σ (`Mathlib.RepresentationTheory.Intertwining`).

Definition 7 (Irreducible representation). `.IsIrreducible`, an abbreviation for `IsSimpleOrder (Subrepresentation )`, says ρ is non-trivial and has no proper non-zero subrepresentation (`Mathlib.RepresentationTheory.Irreducible`). `Representation.irreducible_iff_isSimpleModule_asModule` provides the equivalence with `IsSimpleModule (MonoidAlgebra k G) .asModule`, and the corresponding instance is registered (so simplicity of the $k[G]$ -module is automatic).

Definition 8 (`FDRep k G`). `FDRep k G` (`Mathlib.RepresentationTheory.FDRep`) is the k -linear abelian symmetric monoidal category of finite-dimensional k -linear representations of G . For $V : \text{FDRep } k \text{ } G$ the underlying space is V (via coercion), the action is `V.`, and `[CategoryTheory.Simple V]` encodes simplicity in this category. All the character-theoretic Mathlib lemmas (Section 4.1) are stated in `FDRep`.

Chapter 2

Maschke and Semisimplicity

The whole chapter is essentially the citation `MonoidAlgebra.Submodule.instIsSemisimpleModule`; we just state its consequences explicitly as named statements so subsequent chapters can refer to them.

2.1 Maschke's theorem in Mathlib

Theorem 9 (Maschke: every $k[G]$ -module is semisimple). *Under the standing hypotheses (Definition 1), for every `Module (MonoidAlgebra k G) V` we have `IsSemisimpleModule (MonoidAlgebra k G) V`: every $k[G]$ -submodule has a $k[G]$ -complement.*

Proof. This is the registered instance `MonoidAlgebra.Submodule.instIsSemisimpleModule` in `Mathlib.RepresentationTheory.Maschke`, with hypotheses `[Field k] [Group G] [Finite G] [NeZero (Nat.card G : k)]`. The Mathlib proof uses `LinearMap.equivariantProjection` (averaging π over G) plus `LinearMap.equivariantProjection_condition`. $\square$

Corollary 10 ($k[G]$ is a semisimple ring). *`IsSemisimpleRing (MonoidAlgebra k G)` holds.*

Proof. By definition, `IsSemisimpleRing R := IsSemisimpleModule R R`. Specialise Theorem 9 to $V := \text{MonoidAlgebra } k \ G$. $\square$

Corollary 11 (Every $V : \text{FDRep } k \ G$ is semisimple). *For every $V : \text{FDRep } k \ G$, $V.asModule$ is a semisimple `MonoidAlgebra k G`-module and is finite as a $k[G]$ -module.*

Proof. Direct from Theorem 9; finiteness over $k[G]$ follows from finiteness over k (which is built into `FDRep`) since k is a subring of $k[G]$. $\square$

Chapter 3

Schur's Lemma in Mathlib

Schur is fully formalised in two complementary forms; we just cite both.

3.1 Categorical form (FDRep)

Theorem 12 (Schur, finrank form). *Let $V, W : \text{FDRep } k \ G$ both carry $[\text{CategoryTheory.Simple}]$. Then*

$$\text{finrank}_k(V \rightarrow W) = \begin{cases} 1 & \text{if } V \cong W, \\ 0 & \text{otherwise.} \end{cases}$$

This is `FDRep.finrank_hom_simple_simple` (file `Mathlib.RepresentationTheory.FDRep`).

Proof. Cited directly. The Mathlib proof is the standard one: a non-zero map between simples has trivial kernel and trivial cokernel hence is an isomorphism, so $V \rightarrow W$ is either 0 or $\cong V \rightarrow V$, and on $V \rightarrow V$ algebraic closedness gives an eigenvalue λ , forcing $T - \lambda \text{id} = 0$. $\square$

3.2 Bundled IntertwiningMap form

Theorem 13 (Schur, scalar form). *For ρ irreducible and finite-dimensional over an algebraically closed field k , the canonical algebra map*

$$k \longrightarrow \text{End}_G(\rho) = \rho.\text{IntertwiningMap } \rho$$

is bijective. Hence $\text{End}_G(\rho) \simeq_{k\text{-alg}} k$.

Proof. This is `Representation.IsIrreducible.algebraMap_intertwiningMap_bijective_of_isAlgClosed` (`Mathlib.RepresentationTheory.Irreducible`). $\square$

Theorem 14 (Schur, dimension form). *For ρ as above, $\text{finrank}_k \rho.\text{IntertwiningMap } \rho = 1$.*

Proof. This is `Representation.IsIrreducible.finrank_intertwiningMap_self`. $\square$

Theorem 15 (Schur for distinct irreducibles). *If ρ, σ are both irreducible and there is no G -equivariant isomorphism between them (i.e. `IsEmpty ( .Equiv )`), then `Subsingleton ( .IntertwiningMap )`; equivalently, every intertwiner $\rho \rightarrow \sigma$ is zero.*

Proof. This is the registered instance `Representation.IsIrreducible.instSubsingletonIntertwiningMapOfIsEmpty`. $\square$

Chapter 4

Characters and Their Orthogonality

The character theory we need (definition, conjugation invariance, isomorphism invariance, the average formula, and Schur orthogonality) is fully formalised in `Mathlib.RepresentationTheory.Character`; we list the relevant declarations.

4.1 Mathlib's character API

Definition 16 (Character of $V : \text{FDRep } \mathbf{k} \ G$). $\chi_V(g) = \text{tr}(V \cdot \rho(g))$, defined as `FDRep.character V g`.

Lemma 17 ($\chi_V(1) = \dim V$).

Proof. `FDRep.char_one`. □

Lemma 18 (Class function). $\chi_V(hgh^{-1}) = \chi_V(g)$.

Proof. `FDRep.char_conj`. □

Lemma 19 (Iso-invariance). $V \cong W$ in $\text{FDRep } \mathbf{k} \ G \Rightarrow \chi_V = \chi_W$.

Proof. `FDRep.char_iso`. □

Theorem 20 (Average of χ_V). $|G|^{-1} \sum_{g \in G} \chi_V(g) = \text{finrank}_k V^G$.

Proof. `FDRep.average_char_eq_finrank_invariants`. □

Theorem 21 (Hom-space dimension via characters). $|G|^{-1} \sum_g \chi_V(g) \chi_W(g^{-1}) = \text{finrank}_k (V \rightarrow W)_G$.

Proof. `Representation.card_inv_mul_sum_char_mul_char_eq_finrank`. □

Theorem 22 (Character orthonormality). For $V, W : \text{FDRep } \mathbf{k} \ G$ both simple,

$$|G|^{-1} \sum_g \chi_V(g) \chi_W(g^{-1}) = \begin{cases} 1 & V \cong W, \\ 0 & \text{otherwise.} \end{cases}$$

Proof. `FDRep.char_orthonormal`. □

Chapter 5

Isotypic Decomposition via Mathlib

The full isotypic-component theory is in `Mathlib.RingTheory.SimpleModule.Isotypic`. Specialised to $R := \text{MonoidAlgebra } k \ G$ and the standing hypotheses, every $k[G]$ -module is a sum of its isotypic components, each of which is a finite direct sum of copies of one simple module.

5.1 Mathlib’s isotypic API

Definition 23 (`isotypicComponent`). For a simple R -module S , `isotypicComponent R M S` is the sum of all submodules of M isomorphic to S (`Mathlib.RingTheory.SimpleModule.Isotypic`).

Theorem 24 (Isotypic components are independent and exhaust). *For any `IsSemisimpleModule R M`, the family `isotypicComponents R M` is `sSupIndep` and its `sSup` is $\top$. Combined with `IsIsotypic.linearEquiv_fun`, every isotypic component is isomorphic to S^n for the corresponding simple S and some $n \in \mathbb{N}$.*

Proof. Cited from Mathlib: `sSupIndep_isotypicComponents` and `sSup_isotypicComponents` (in `Mathlib.RingTheory.SimpleModule.Isotypic`); the per-component description is `IsIsotypic.linearEquiv_fun`. $\square$

Corollary 25 (Module-form Peter–Weyl, Mathlib version). *For any `IsSemisimpleModule R M` with `Module.Finite R M`, there exist $n \in \mathbb{N}$, simple submodules $(S_i)_{i < n}$ of M , and dimensions $(d_i)_{i < n}$ with each $d_i \neq 0$, such that*

$$\text{End}_R M \simeq_{R_0\text{-alg}} \prod_{i < n} \text{Mat}_{d_i \times d_i}(\text{End}_R S_i).$$

Proof. This is `IsSemisimpleModule.exists_end_algEquiv_pi_matrix_end` in `Mathlib.RingTheory.SimpleModule`. $\square$

Chapter 6

The Peter–Weyl Theorem for Finite Groups

This is now a one-step application of Wedderburn–Artin to the regular representation, plus an algebraic-closedness step to identify the division algebras with k .

6.1 The regular representation

Definition 26 (Left regular representation). `Representation.leftRegular k G : Representation k G ((G → k))`. Its bundled form is `Rep.ofMulAction k G G : Rep k G`, with the same underlying $k[G]$ -module as `MonoidAlgebra k G` acting on itself by left multiplication.

Lemma 27 ($\chi_{k[G]}$ is supported at 1). $\chi_{k[G]}(g) = |G|$ if $g = 1$ and 0 otherwise.

Proof. In the basis $\{\delta_h : h \in G\}$, the matrix of left translation by g is the permutation matrix for $h \mapsto gh$. Its trace counts fixed points: $h = gh$ iff $g = 1$, in which case all $|G|$ basis vectors are fixed. $\square$

6.2 Wedderburn form of $k[G]$

Theorem 28 (Wedderburn–Artin for $k[G]$). *Under the standing hypotheses there exist $n \in \mathbb{N}$, division k -algebras $(D_i)_{i < n}$ each finite over k , and $(d_i)_{i < n}$ with each $d_i \neq 0$, together with a k -algebra isomorphism*

$$\text{MonoidAlgebra } k \ G \simeq_{k\text{-alg}} \prod_{i < n} \text{Mat}_{d_i \times d_i}(D_i).$$

Proof. Apply `IsSemisimpleRing.exists_algEquiv_pi_matrix_divisionRing_finite` with $R_0 := k$, $R := \text{MonoidAlgebra } k \ G$. The hypothesis `[Module.Finite k (MonoidAlgebra k G)]` holds because `MonoidAlgebra k G` has k -dimension $|G| < \infty$. The hypothesis `[IsSemisimpleRing R]` is Corollary 10. $\square$

6.3 Algebraic closedness collapses the D_i to k

Lemma 29 (Finite-dim division algebra over alg-closed k equals k). *Let k be an algebraically closed field and D a finite-dimensional division k -algebra. Then the canonical map $k \rightarrow D$ is a*

k-algebra isomorphism.

Proof. The canonical map $\iota : k \rightarrow D$ is injective (it has trivial kernel since k is a field and $\iota(1) = 1 \neq 0$). For surjectivity, take $d \in D$. Since D is finite-dimensional over k , the powers $1, d, d^2, \dots$ are k -linearly dependent, so d satisfies some non-zero polynomial $f \in k[X]$. Pick f of minimal degree; it is then irreducible in $k[X]$ (else a proper factorisation would give zero-divisors in D , contradicting D a division ring). Since k is algebraically closed, every irreducible polynomial in $k[X]$ has degree 1, so $\deg f = 1$ and $d \in \iota(k)$. $\square$

Theorem 30 (Peter–Weyl for finite groups, ring form). *Under the standing hypotheses there exist $n \in \mathbb{N}$ and $(d_i)_{i < n}$ with each $d_i \neq 0$, together with a k -algebra isomorphism*

$$\text{MonoidAlgebra } k \ G \simeq_{k\text{-alg}} \prod_{i < n} \text{Mat}_{d_i \times d_i}(k).$$

Proof. Apply Theorem 28 to obtain $\Phi : k[G] \simeq_k \prod_i \text{Mat}_{d_i}(D_i)$ with each D_i a finite-dimensional division k -algebra. Apply Lemma 29 componentwise to obtain $D_i \simeq k$ as k -algebras, and apply the matrix-algebra functor to get $\text{Mat}_{d_i}(D_i) \simeq \text{Mat}_{d_i}(k)$. Compose. $\square$

6.4 Sum-of-squares dimension formula

Corollary 31 (Sum of squares of dimensions). *Under the standing hypotheses, with $(d_i)_i$ the dimensions appearing in Theorem 30,*

$$|G| = \sum_{i < n} d_i^2.$$

Proof. Take $\dim_k$ of both sides of the isomorphism in Theorem 30: the left side is $\dim_k k[G] = |G|$; the right side is $\sum_i \dim_k \text{Mat}_{d_i}(k) = \sum_i d_i^2$. $\square$

6.5 Module-form Peter–Weyl

Theorem 32 (Module form of Peter–Weyl). *Under the standing hypotheses there exist $n \in \mathbb{N}$, simple subrepresentations $(V_i)_{i < n}$ of the regular representation, and multiplicities $(d_i)_{i < n}$ with each $d_i = \dim_k V_i \neq 0$, such that*

$$\text{MonoidAlgebra } k \ G \simeq \bigoplus_{i < n} V_i^{\oplus d_i}$$

as $k[G]$ -modules. *Equivalently: every irreducible appears in $k[G]$ with multiplicity equal to its k -dimension, and these are all the irreducibles.*

Proof. Apply Corollary 25 to $R := \text{MonoidAlgebra } k \ G$, $R_0 := k$, and $M := \text{MonoidAlgebra } k \ G$ (a finite $k[G]$ -module since $|G| < \infty$ and so a finite k -module). This produces simple subrepresentations V_i and dimensions d_i together with a k -algebra isomorphism $\text{End}_{k[G]} k[G] \simeq \prod_i \text{Mat}_{d_i}(\text{End}_{k[G]} V_i)$. Now $\text{End}_{k[G]} k[G] \simeq k[G]^{\text{op}}$ canonically (`moduleEndSelf = MulOpposite.opAlgEquiv`), and by Schur (Theorem 13) $\text{End}_{k[G]} V_i \simeq k$. The isomorphism then becomes $k[G]^{\text{op}} \simeq \prod_i \text{Mat}_{d_i}(k)$, recovering Theorem 30 (and its module form by reading off the decomposition of the standard $k[G]$ -module on the right). The dimension identification $d_i = \dim_k V_i$ comes from comparing dimensions of $V_i^{\oplus d_i}$ and $\text{Mat}_{d_i}(k)$ via the equivalence $\text{Mat}_{d_i}(k) \cong V_i^{\oplus d_i}$ as left $\text{Mat}_{d_i}(k)$ -modules. $\square$

Chapter 7

Character-Theoretic Consequences

These follow immediately from the Mathlib character API combined with the isotypic decomposition.

7.1 Multiplicity formula and irreducibility test

Theorem 33 (Multiplicity formula). *Let $V : \mathbf{FDRep} \ \mathbf{k} \ G$ and ξ a simple $V_\xi : \mathbf{FDRep} \ \mathbf{k} \ G$. Write m_ξ for the multiplicity of V_ξ in the isotypic decomposition of V (Theorem 24). Then*

$$m_\xi = |G|^{-1} \sum_g \chi_V(g) \chi_{V_\xi}(g^{-1}) = \text{finrank}_k(V_\xi \rightarrow V)_G.$$

Proof. Decompose $V \cong \bigoplus_\eta V_\eta^{\oplus m_\eta}$ via Theorem 24. Then $\text{finrank}_k(V_\xi \rightarrow V)_G = \sum_\eta m_\eta \text{finrank}_k(V_\xi \rightarrow V_\eta)_G = m_\xi$ by Theorem 12. The first equality is then Theorem 21. $\square$

Corollary 34 (Irreducibility test). *$V : \mathbf{FDRep} \ \mathbf{k} \ G$ is simple iff $|G|^{-1} \sum_g \chi_V(g) \chi_V(g^{-1}) = 1$.*

Proof. Decompose $V \cong \bigoplus_\xi V_\xi^{\oplus m_\xi}$ and expand. By Theorem 22, $|G|^{-1} \sum_g \chi_V(g) \chi_V(g^{-1}) = \sum_\xi m_\xi^2$. This integer is 1 iff exactly one m_ξ is 1 and the rest are 0, iff $V \cong V_\xi$ is simple. $\square$

Corollary 35 (Characters separate isomorphism classes). *$V \cong W$ in $\mathbf{FDRep} \ \mathbf{k} \ G$ iff $\chi_V = \chi_W$.*

Proof. $\Rightarrow$: Lemma 19. $\Leftarrow$: by Theorem 33 the multiplicities $m_\xi(V)$ and $m_\xi(W)$ depend only on the characters, so they coincide; uniqueness of the isotypic decomposition (Theorem 24) then forces $V \cong W$. $\square$

7.2 Class functions and the second orthogonality

Definition 36 (Class functions). $\text{Cl}(G) := \{f : G \rightarrow k \mid \forall g, h. f(hgh^{-1}) = f(g)\}$, a k -subspace of k^G .

Lemma 37 ($\dim \text{Cl}(G) = \#\text{conjugacy classes}$). $\dim_k \text{Cl}(G) = |\text{ConjClasses } G|$.

Proof. The indicator functions of conjugacy classes form a basis of $\text{Cl}(G)$. $\square$

Theorem 38 (Irreducible characters span $\text{Cl}(G)$). *The set $\{\chi_V : V \in \text{FDRep } k \ G, \text{ Simple } V\}$ spans $\text{Cl}(G)$ over k .*

Proof. Under Theorem 30 the centre of $k[G]$ pulls back to the centre of $\prod_i \text{Mat}_{d_i}(k)$, which is $\prod_i k$, of k -dimension n (the number of simples). On the other hand the centre of $k[G]$ identifies with $\text{Cl}(G)$ via $\sum_g a_g g \leftrightarrow (g \mapsto a_g)$ (an element commutes with every $h \in G$ iff its support function is conjugation-invariant). Hence $\dim_k \text{Cl}(G) = n$, the number of simples. Since the irreducible characters are orthonormal (Theorem 22), there are n of them, and they live in $\text{Cl}(G)$ by Lemma 18; hence they span (a linearly-independent set of size equal to the dimension is a basis). $\square$

Theorem 39 (Number of irreducibles = number of conjugacy classes). *The number of isomorphism classes of simple objects of $\text{FDRep } k \ G$ equals $|\text{ConjClasses } G|$.*

Proof. By Theorem 38 and Theorem 22, the irreducible characters form a basis of $\text{Cl}(G)$; their number is therefore $\dim_k \text{Cl}(G) = |\text{ConjClasses } G|$ (Lemma 37). $\square$

Chapter 8

Mathlib-Status Summary

Already in Mathlib ([ML], with \leanok). The following declarations are direct citations: `Representation`, `Representation.asModule`, `Subrepresentation` (and its order isomorphism with $k[G]$ -submodules), `Representation.IntertwiningMap`, `Representation.Equiv`, `Representation.IsIrreducible` (and the equivalence with `IsSimpleModule`), `FDRep`, `Representation.leftRegular`, `Rep.ofMulAction`, `MonoidAlgebra.Submodule.instIsSemisimpleModule` (Maschke), `IsSemisimpleRing` as a derived instance, `FDRep.finrank_hom_simple_simple` (Schur), `Representation.IsIrreducible.algebraMap_intertwining` (scalar Schur), `Representation.IsIrreducible.finrank_intertwiningMap_self` (dimension Schur), `Representation.IsIrreducible.instSubsingletonIntertwiningMapOfIsEmptyEquiv` (Schur for distinct irreducibles), `FDRep.character` and the basic identities `FDRep.char_one`, `FDRep.char_conj`, `FDRep.char_iso`, `FDRep.average_char_eq_finrank_invariants`, `Representation.card_inv_m`, `FDRep.char_orthonormal`, `isotypicComponent`, `IsIsotypic`, `sSupIndep_isotypicComponents`, `sSup_isotypicComponents`, `IsIsotypic.linearEquiv_fun`, `IsSemisimpleModule.exists_end_algEquiv_pi_matrix`, `IsSemisimpleRing.exists_algEquiv_pi_matrix_divisionRing_finite`.

Trivial wrappers ([~ML]). `FDRep.isSemisimpleModule_asModule` (Corollary 11), `FDRep.character_leftRegular` (Lemma 27), `Representation.multiplicity_eq_inner_char` (Theorem 33), `FDRep.irreducible_iff_inner_self` (Corollary 34), `FDRep.iso_iff_character_eq` (Corollary 35), `ClassFunction` and `ClassFunction.finrank_eq_conjClasses`.

Genuinely new contributions ([NEW]). There are only four:

1. `IsAlgClosed.divisionRing_algEquiv` — every finite-dimensional division algebra over an algebraically closed field equals the field (Lemma 29). The proof is a 5-line argument.
2. `PeterWeyl.groupAlgebra_algEquiv_pi_matrix` (Theorem 30). Combines the Mathlib Wedderburn–Artin output with the previous lemma; one line of `simp` and apply.
3. `PeterWeyl.sum_sq_dim_eq_card` (Corollary 31). `finrank` of both sides.
4. `FDRep.span_irreducibleCharacters_eq_top` and the corollary `FDRep.num_simple_eq_num_conjClasses` (Theorems 38, 39). The argument identifies the centre of $k[G]$ with $\text{Cl}(G)$ and reads off dimension from Theorem 30.

What this means. The whole project is essentially a five-page Lean file: `Maschke + WedderburnArtin + algClosed division-algebra collapse`. The proof of Peter–Weyl in this blueprint sidesteps the classical isotypic calculation entirely (which is itself already in Mathlib) and instead routes through the ring-theoretic Wedderburn–Artin theorem.